

Lesson 6: Theoretical Foundations of SHA-256

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Lecture Goals

This lecture introduces:

- What cryptographic hash functions are,
- The purpose of SHA-256,
- How SHA-256 processes data,
- The main stages of the algorithm,
- Important security properties of SHA-256.

1 Introduction

SHA-256 stands for:

Secure Hash Algorithm 256-bit

It belongs to the SHA-2 family of cryptographic hash functions. SHA-256 converts data of any length into a fixed-size output of 256 bits.

2 What Is a Hash Function?

A hash function transforms input data into a shorter representation called a hash.

```
Input:
Hello World

Output:
a591a6d40bf42040...
```

The output always has the same length regardless of the input size.

3 Properties of Cryptographic Hash Functions

A secure hash function should satisfy:

- Determinism,
- Preimage resistance,
- Second preimage resistance,
- Collision resistance,
- Avalanche effect.

4 Determinism

The same input always produces the same output.

```
SHA-256("Hello")  
=  
SHA-256("Hello")
```

The result never changes.

5 Preimage Resistance

Given a hash value:

```
2cf24dba5fb0...
```

it should be computationally infeasible to determine the original message. This property makes SHA-256 a one-way function.

6 Collision Resistance

A collision occurs when:

```
Message A != Message B
```

but

```
SHA-256(A) = SHA-256(B)
```

A secure hash function makes collisions extremely difficult to find.

7 Avalanche Effect

A tiny change in the input should completely change the output.

Example:

```
hello  
Hello
```

These two messages produce completely different hashes. This property prevents attackers from predicting relationships between inputs and outputs.

8 Overview of SHA-256

SHA-256 processes data in blocks of:

512 bits

Each block is processed through multiple rounds of mathematical operations. The final result is always:

256 bits

long.

9 Step 1: Padding

Before processing begins, the message must be padded.

Padding adds:

- One bit equal to 1,
- Multiple bits equal to 0,
- The original message length.

The goal is to make the total length a multiple of 512 bits.

10 Step 2: Splitting into Blocks

After padding, the message is divided into blocks.

```
Block 1 : 512 bits  
Block 2 : 512 bits  
Block 3 : 512 bits  
...
```

Each block is processed independently.

11 Step 3: Initial Hash Values

SHA-256 starts with eight predefined 32-bit values. These values are called:

```
H0 , H1 , H2 , H3 ,  
H4 , H5 , H6 , H7
```

They are fixed constants defined in the standard.

12 Step 4: Message Schedule

Each 512-bit block is expanded into:

```
W0 , W1 , W2 , ...  
W63
```

This creates 64 words of 32 bits each. Additional words are generated using bitwise operations.

13 Bitwise Operations

SHA-256 relies heavily on:

- XOR,
- AND,
- NOT,
- Bit shifts,
- Bit rotations.

Example XOR:

```
1 XOR 1 = 0
1 XOR 0 = 1
0 XOR 1 = 1
0 XOR 0 = 0
```

These operations are efficient and easy for processors to execute.

14 Step 5: Compression Function

The core of SHA-256 is the compression function. Eight working variables are used:

```
a b c d e f g h
```

The algorithm performs:

64 rounds

of mixing operations. Each round modifies the variables.

15 Choice Function

One important function is:

$$Ch(x, y, z) = (x \wedge y) \oplus (\neg x \wedge z)$$

It selects bits from either:

- y,
- z,

depending on x.

16 Majority Function

Another function is:

$$Maj(x, y, z)$$

which returns the majority value of the three inputs. Examples:

```
1 1 0 -> 1
1 0 0 -> 0
```

This increases diffusion inside the algorithm.

17 Step 6: Updating the Hash State

After every round:

```
a b c d e f g h
```

are updated. After all 64 rounds, the result is added to:

```
H0 ... H7
```

The process repeats for every message block.

18 Step 7: Final Hash

When all blocks have been processed:

```
H0 || H1 || H2 || H3 ||
H4 || H5 || H6 || H7
```

are concatenated.

The final output contains:

256 bits

or

64 hexadecimal characters.

19 Applications of SHA-256

SHA-256 is widely used in:

- Digital signatures,
- Blockchain systems,
- File integrity verification,
- TLS certificates,
- Software distribution.

20 Limitations

SHA-256 is designed for hashing data. It is not ideal for password storage because it is very fast. Modern systems prefer:

- bcrypt,
- PBKDF2,
- scrypt,
- Argon2.

These algorithms intentionally require more computation.

Summary

In this lecture, we covered:

- Cryptographic hash functions,
- Security properties of SHA-256,
- Padding,
- Message blocks,
- Message schedules,
- Compression rounds,
- Applications and limitations of SHA-256.

21 Exercises

1. What is the purpose of a cryptographic hash function?
2. Explain the difference between:
 - Preimage resistance,
 - Collision resistance.
3. What is the avalanche effect?
4. Why does SHA-256 use padding?
5. How large is a SHA-256 output?
6. How many rounds are performed for each block?
7. Explain the role of:
 - XOR,

- AND,
 - Bit rotations.
8. Why is SHA-256 considered a one-way function?
 9. List three real-world applications of SHA-256.
 10. Challenge:
Draw a diagram showing the complete SHA-256 workflow from the original message to the final 256-bit hash.